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EDUCATION

Xi'an Jiaotong-Liverpool University (XJTLU) / Suzhou, China

Sep 2021 - Jul 2025

Bachelor of Engineering with Honors in Electronic Science and Technology

First Class (70.5%)

The University of Manchester / Manchester, UK

Sep 2025 - Sep 2026

Master of Science in Robotics, expected to be awarded on Sep 2026

Relevant Coursework:

Robotic Systems; Software for Robotics; Cognitive Robotics and Computer Vision;
Autonomous Mobile Robots; Reinforcement Learning

Skills to be Developed:

ROS 2, Gazebo, Linux, Reinforcement Learning for control, Computer Vision algorithms

LANGUAGE PROFICIENCY & TECHNICAL SKILLS

Native Speaker of Chinese Mandarin;

English: **C1** (IELTS: 7.0; L:7.0, R:7.5, W:6.5, S:6.0).

Proficient:

Programming: Python (OpenCV, ML libraries), C/C++, MATLAB

EDA & Simulation: PLECS, LTspice, Quartus, CST Microwave Studio

INTERNSHIP EXPERIENCE

Yancheng Sanfeng Machinery Co., Ltd.

Mar 2025 - Apr 2025

Field: Manufacture

Position: Intelligent Equipment Transformation Intern

Participated in the intelligent transformation of mechanical equipment, involving the design of automation control systems and the development of remote monitoring solutions.

Key Contribution:

- Developed a remote monitoring system prototype based on digital twin technology to enable equipment connectivity and networking;
- Assisted in the electrical design of mechanical equipment automation systems and supported equipment testing and performance optimization;

Outcome:

- Achieved 95% real-time data accuracy in equipment status monitoring, contributing to the foundation of the company's digital transformation;
- Enhanced the technical team's efficiency in data analysis and system optimization.

RESEARCH EXPERIENCE

Application of Digital Twin in Monitoring Power Electronic Converters

Jul 2024 - Apr 2025

Individual Researcher, Final Year Project

Advisor: Prof. Xuchen Wang, Assistant Professor

(Department of Electrical and Electronic Engineering, XJTLU)

Personal Research Content:

- Built a high-fidelity digital replica of a buck converter using **MATLAB (Simulink)**, implementing system dynamics via **Fourth-order Runge-Kutta numerical integration**;
- Applied **Particle Swarm Optimization (PSO)** for parameter identification and state estimation under both open-loop and closed-loop configurations.

Outcome:

- Developed a non-invasive condition monitoring system for power electronic converters based on a **digital twin framework**, enhancing reliability and predictive maintenance capabilities;
- Achieved robust and cost-effective condition monitoring with potential applications in robotic drive

systems and industrial power management.

Buck DC-DC Converter with Analog Control

Jul 2024 - Sep 2024

Undergraduate Research Assistant, XJTU

Advisor: Prof. Xuchen Wang, Assistant Professor

(Department of Electrical and Electronic Engineering, XJTU)

Personal Duty:

Designed and stabilized a DC-DC buck converter with analog feedback control for precise voltage regulation;

Utilized **frequency-domain analysis** and **Bode plots** to design a **Type III compensator** for loop stability;

Simulated transient and steady-state performance under varying loads using **PLECS** and **MATLAB**.

Outcome: Delivered a **stable 5Vdc output from a 24Vdc input**, demonstrating strong analog circuit design and control tuning skills relevant to embedded systems and power electronics.

Compact Dual-Polarized Antenna with Meta-material Backing for Indoor /Outdoor WPAN Applications

Jul 2023 - Aug 2023

Undergraduate Research Assistant, XJTU

Advisor: Prof. Zhenzhen Jiang, Assistant Professor

(Department of Communications and Networking, XJTU)

Objective: To design a wearable antenna with an electromagnetic band-gap (EBG) ground plane made of textile materials to work at 2.45GHz in the industrial, scientific, and medical radio band for Bluetooth Low Energy (BLE) applications.

Main Duties:

- Employed **CST Microwave Studio** for EM simulation and SAR analysis to ensure human safety compliance;
- Optimized meta-material unit cells to enhance antenna gain and reduce backward radiation.

Outcome: Developed a smart belt antenna with textile-based EBG structure, operating at 2.45 GHz with improved efficiency and radiation patterns for WPAN applications.

PROJECTS/AWARDS

The 2024 Electrical & Electronic Engineering Innovation Competition

Dec 2023 - Jul 2024

Project Details: Dual-Active Bridge (DAB) DC-DC Converter

(Schneider Electric Go Green Power Electronics Innovation Track)

Awarded Provincial Third Prize in the 2024 Electrical & Electronic Engineering Innovation Competition

Supervisor: Prof. Xuchen Wang, Assistant Professor; Prof. Wen Liu, Senior Associate Professor

(Department of Electrical and Electronic Engineering, XJTU)

Role: Core Team Member & Hardware Design Lead

Main Duty:

Designed and prototyped a **1kW, 36-60V input to 400V output DAB converter**, specializing in high-frequency transformer/inductor design and system analysis to achieve high efficiency and power density.

Key Contributions:

- Conducted in-depth analysis of converter topologies and phase-shift modulation principles, selecting the **DAB** for its bidirectional power flow and soft-switching operation;
- Spearheaded the magnetic component design: engineered a high-frequency transformer (**5:56 ratio**) and inductor, optimizing **core geometric constants (Kg)**, **wire gauges**, and **leakage inductance** to balance performance and thermal management;
- Designed and integrated voltage/current sensing circuits with appropriate **filter networks and ADC interfaces** for accurate feedback control;
- Validated converter performance through **MATLAB/PLECS simulations** and hardware testing, confirming efficiency metrics and stable operation under load;
- Performed a comprehensive **economic analysis** of components, identifying cost-effective alternatives to reduce overall project cost without compromising quality.